

# run\_sjsdm\_tuning\_candidates.R

August 12, 2026

## Contents

|                                       |          |
|---------------------------------------|----------|
| run_sjsdm_tuning_candidates . . . . . | 1        |
| <b>Index</b>                          | <b>3</b> |

---

|                                    |
|------------------------------------|
| run_sjsdm_tuning_candidates        |
| <i>Run sjSDM Tuning Candidates</i> |

---

## Description

Fits every deterministic regularization candidate independently on each cross-validation training partition and scores held-out probabilities.

## Usage

```
run_sjsdm_tuning_candidates(  
  data_assignments = NULL,  
  data_candidates = NULL,  
  prepare_fold_function = NULL,  
  fit_function = NULL,  
  predict_function = NULL,  
  score_function = score_sjsdm_tuning_predictions,  
  seed = 900723L,  
  epsilon = 1e-06,  
  retain_prediction_cache = FALSE  
)
```

## Arguments

- data\_assignments**  
Location-level assignment table with ‘repeat\_id’, ‘fold\_id’, ‘location\_id’, ‘n\_samples’, and ‘row\_indices’. Optional ‘cv\_strategy’ is propagated.
- data\_candidates**  
Candidate table returned by [build\_sjsdm\_regularization\_candidates()].
- prepare\_fold\_function**  
Injectable function called with ‘train\_indices’, ‘test\_indices’, ‘repeat\_id’, and ‘fold\_id’. It must return ‘data\_train\_input’, ‘data\_test\_input’, and matrix ‘data\_test\_observed’.

|                                      |                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fit_function</code>            | Injectable function called with <code>'data_train_input'</code> , one-row <code>'candidate'</code> , and deterministic integer <code>'seed'</code> .                                                                                                                                                         |
| <code>predict_function</code>        | Injectable function called with fitted <code>'object'</code> and <code>'data_test_input'</code> . It must return a probability matrix aligned to <code>'data_test_observed'</code> .                                                                                                                         |
| <code>score_function</code>          | Injectable function called with the fitted object, test input, aligned observations and probabilities, <code>epsilon</code> , and, when supported, a deterministic <code>'score_seed'</code> . Defaults to marginal binary prediction scoring; production sjSDM tuning supplies the joint-likelihood scorer. |
| <code>seed</code>                    | Single non-negative integer used to derive stable fit and score seeds. Defaults to <code>'900723L'</code> .                                                                                                                                                                                                  |
| <code>epsilon</code>                 | Probability clipping tolerance used for held-out log loss. Defaults to <code>'1e-6'</code> .                                                                                                                                                                                                                 |
| <code>retain_prediction_cache</code> | Logical. When true, retain compact fold metadata and candidate probability matrices so selected OOF artifacts can be assembled without refitting.                                                                                                                                                            |

### Details

Fold preparation is run once per repeat and fold. Preparation, fit, prediction, and scoring errors are retained as structured status rows so a failed candidate does not abort the remaining tuning grid.

### Value

Compact tibble with one row per repeat, fold, and candidate. It contains candidate parameters, fold counts, total and per-response held-out negative log likelihood, macro AUC, fit status, error text, CV strategy, and regularization source. Fitted objects and prediction matrices are omitted by default. With `'retain_prediction_cache = TRUE'`, a named list contains the unchanged tuning table and one compact cache entry per fold.

# Index

run\_sjsdm\_tuning\_candidates, [1](#)